

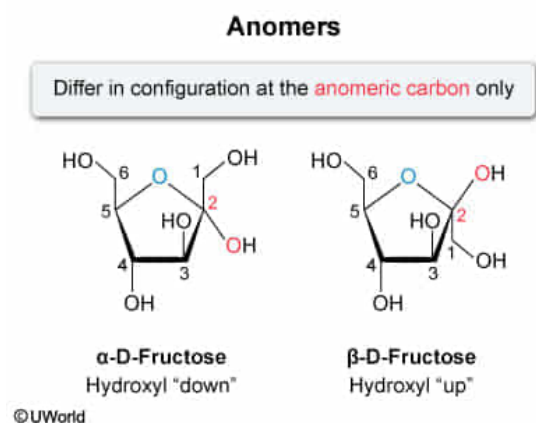
Can α -D-fructose and β -D-fructose be classified as anomers?

- ✔ ☒ A. Yes, because they are ketoses that differ in configuration at C2 only. [37%]
☐ B. Yes, because they differ in configuration at more than one carbon. [5%]
☐ C. Yes, because they are C1 epimers. [49%]
☐ D. No, because they are enantiomers. [7%]

Correct

38% answered correctly

Explanation:



Stereoisomers are molecules that have the same atom connectivity but differ in the spatial orientation of the atoms. Stereoisomers contain one or more **stereocenters**, or **chiral centers**, which consist of an atom (typically carbon) with four different substituents. Diastereomers, a type of stereoisomer, have more than one stereocenter and differ in orientation at some, but not all, stereocenters.

Some **sugars** can be classified as a special **kind of diastereomer**, known as an **anomer**, that **differs in configuration at only the anomeric carbon**. The **α** and **β** designations in a sugar's name refer to different orientations of the hydroxyl group at the anomeric carbon.

Carbon atoms in sugars are numbered beginning with the carbon that is closest to the anomeric carbon when the sugar is in linear form. For **aldoses** such as glucose, the anomeric carbon is C1, whereas for **ketoses** such as fructose, the anomeric carbon is C2. α -D-fructose and β -D-fructose are diastereomers that differ in orientation at C2 (the anomeric carbon) only. Therefore, α -D-fructose and β -D-fructose are anomers.

(Choice B) Anomers differ in configuration at only *one* position (the anomeric carbon), not multiple positions.

(Choice C) **Epimers** are diastereomers that differ in configuration at one position, and anomers are a specific type of epimer that differs at the anomeric carbon. However, for fructose, the anomeric carbon is C2, not C1.

(Choice D) α -D-fructose and β -D-fructose are *not* **enantiomers** because they differ in configuration at *only one* stereocenter, whereas enantiomers differ in configuration at *every*

stereocenter. The enantiomer of α -D-fructose is α -L-fructose, and the enantiomer of β -D-fructose is β -L-fructose.

Educational objective:

Stereoisomers are molecules containing the same atom connectivity but different spatial orientation of the atoms. Sugar stereoisomers that differ in orientation at only one stereocenter are known as epimers, a specific kind of diastereomer. If the sugars differ in configuration at only the anomeric carbon (denoted by α and β in the sugar's name), they are classified as anomers, a special kind of epimer.

Time spent:0 sec

Subject:
Organic Chemistry

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System:
Introduction to Organic Chemistry